Task 2: Firewall Configuration & System Hardening

1. Objective

Enhance system security by configuring a local firewall (UFW) on a Kali Linux VM to restrict unauthorized access, allow only essential services, disable unnecessary services, apply baseline hardening, and verify the rules by testing access to restricted ports from a remote machine.

2. Lab Setup

Item	Details
Target machine	Kali Linux VM (Oracle VirtualBox)
Test machine (remote)	Windows 11 host laptop, PowerShell
Adapter 1	NAT: `eth0`, 10.0.2.15, gateway 10.0.2.2 (internet access)

Item	Details
Adapter 2	Host-only: `eth1`, 192.168.187.6/24
Windows host IP (host-only)	192.168.187.3
Firewall	UFW (Uncomplicated Firewall)
Tools	UFW, netcat (`nc`), `ss`, `systemctl`, PowerShell `Test-NetConnection`

3. Steps Performed

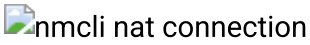
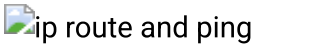
3.1 Network setup and verification

`eth0` (NAT) had no connection profile, so the VM had no default route and no internet. A profile was created and activated:

```
sudo nmcli connection add type ethernet
ifname eth0 con-name nat autoconnect yes
sudo nmcli connection up nat
nmcli device status
ip route
ping -c 3 8.8.8.8
```

Result: both adapters show as connected, the default route is `default via 10.0.2.2 dev eth0`, and internet access works.

PoC: `01-nmcli-nat-connection.png`, `02-ip-route-ping.png`

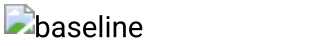

nmcli nat connection

ip route and ping

3.2 Baseline before the firewall

```
sudo ufw status
sudo ss -tulpn
```

Result: UFW is `inactive`, and `ss -tulpn` shows no listening TCP/UDP ports.

PoC: `03-baseline-ufw-inactive-ss.png`

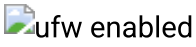

baseline

3.3 Install and configure UFW

```
sudo apt install ufw -y
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow from 192.168.187.0/24 to
any port 22 proto tcp
sudo ufw enable
sudo ufw status verbose
```

Result: UFW is active and enabled on startup. Default policy is deny (incoming) and allow (outgoing). The SSH rule is a precaution (restricted to the host-only subnet). SSH is not running on the VM, and the rule is removed in step 3.6.

PoC: `04-ufw-enabled-status.png`



3.4 Firewall rule testing (local and remote)

A test listener was started on Kali with `nc -lvnp 8080`. Connections were attempted from the Windows host with:

```
Test-NetConnection 192.168.187.6 -Port 8080
```

Test	Firewall state	Result
1. Restricted port	UFW active, default deny, no rule for 8080	`TcpTestSucceeded : False`, "TCP connect failed" (traffic dropped). `PingSucceeded : True` because UFW allows ICMP echo by default.

Test	Firewall state	Result
2. Firewall disabled (control)	`sudo ufw disable`	Kali printed `connect to [192.168.187.6] from (UNKNOWN) [192.168.187.3] 31318`. Port 8080 is reachable.
3. Allow rule added	`sudo ufw allow from 192.168.187.0/24 to any port 8080 proto tcp`	`TcpTestSucceeded : True`. Connection accepted.

Conclusion: the block in Test 1 is caused by the UFW default-deny policy, because the same port is reachable when UFW is disabled (Test 2) or when an explicit allow rule exists (Test 3).

PoC: `05-test1-port8080-blocked-windows.png`, `06-test2-firewall-disabled-connection-received.png`, `07-test3-allow-rule-port8080-success.png`


test 1 blocked

test 2 firewall disabled

test 3 allow rule

3.5 Disable unnecessary services and remote access

Running services were reviewed:

```
systemctl list-units --type=service --
state=running
sudo ss -tulpn
```

Result: no remote-access or network services (SSH, web server, database, Samba, RDP) were running, and no ports were listening. Services not needed on a VM were masked:

```
sudo systemctl mask --now ModemManager
colord
```

PoC: `08-running-services-before.png`, `09-ss-no-listening-ports.png`, `10-services-after-mask.png`

 services before
 no listening ports
 services after

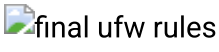
3.6 System hardening and cleanup

- [ ] Removed unneeded rules (SSH is not running, so no ports need to be open):

`sudo ufw delete allow from 192.168.187.0/24 to any port 22 proto tcp`

- [ ] Updated the system: `sudo apt update && sudo apt full-upgrade -y`
- [ ] Set a strong password: `passwd`
- [ ] Locked direct root password login: `sudo passwd -l root`
- [ ] Enabled firewall logging: `sudo ufw logging medium`

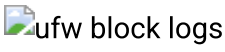
PoC: `11-ufw-final-numbered.png`


final ufw rules

3.7 Final verification

- [ ] From Windows, ports 22 and 8080 both return `TcpTestSucceeded : False`.
- [ ] Blocked attempts are visible in the kernel log: `sudo journalctl -k | grep "UFW BLOCK"`

PoC: `12-final-ports-22-8080-blocked.png`, `13-ufw-block-logs.png`


final ports blocked

ufw block logs

4. PoC Checklist

#	Evidence	Screenshot file
1	NAT profile created, adapters connected	`01-nmcli-nat-connection.png`
2	Default route via eth0, internet ping works	`02-ip-route-ping.png`
3	Baseline: UFW inactive, no listening ports	`03-baseline-ufw-inactive-ss.png`
4	UFW enabled, default deny incoming	`04-ufw-enabled-status.png`
5	Test 1: port 8080 blocked from Windows	`05-test1-port8080-blocked-windows.png`
6	Test 2: firewall disabled, connection received	`06-test2-firewall-disabled-connection-received.png`
7	Test 3: allow rule added,	`07-test3-allow-rule-port8080-success.png`

#	Evidence	Screenshot file
	connection succeeds	
8	Running services list (before)	`08-running-services-before.png`
9	`ss -tulpn` with no listening ports	`09-ss-no-listening-ports.png`
10	Running services list (after masking)	`10-services-after-mask.png`
11	Final UFW rules (numbered)	`11-ufw-final-numbered.png`
12	Final test: ports 22 and 8080 blocked	`12-final-ports-22-8080-blocked.png`
13	UFW BLOCK entries in the log	`13-ufw-block-logs.png`

5. Conclusion

The Kali VM now runs UFW with a default-deny incoming policy and no unnecessary listening services. Access to a restricted port (8080) was

blocked from a remote machine, reachable only when the firewall was disabled or an explicit allow rule was added. This confirms the firewall rules work as intended.